1. CTEs: WITH

- A named step, like a sticky note
- WITH name AS (…) then use it like a table
- Makes big queries easy to read

```
WITH long_books AS (SELECT title, pages
FROM books WHERE pages >= 300)
SELECT * FROM long_books;
```

title	pages
Harry Potter	309
Wonder	310
The Hobbit	310

2. Multiple CTEs

- Separate steps with commas
- Only the first needs WITH
- Later steps can use earlier ones

3. Recursive CTEs

- Start row, then UNION ALL the next row
- A WHERE condition says stop
- Great for counting and calendars

```
WITH RECURSIVE n(x) AS (SELECT 1 UNION ALL
SELECT x + 1 FROM n WHERE x < 5)
SELECT x FROM n;
```

x
1
2
3
4
5

4. Correlated subqueries & EXISTS

- Correlated: runs again for every row
- EXISTS: at least one match?
- NOT EXISTS: no match at all

```
SELECT title FROM books AS b WHERE NOT EXISTS
(SELECT 1 FROM loans AS l
WHERE l.book_id = b.id);
```

title
Charlotte's Web
Coraline

5. Practice free + what's next

- sqliteonline.com, db-fiddle.com, sqlime.org (choose SQLite)
- sqlbolt.com for free lessons
- Paste the setup from the practice guide, click Run
- Next, Part 8: views, indexes, transactions